My Interest in HEC-RAS

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HEC-RAS (short for Hydrologic Engineering Center River Analysis System) is a simulation software used to model the hydraulics of water flow through natural rivers and other open channels. I am a fan of HEC-RAS because it connects hydraulic theory with practical flood modeling. This sample introduces the Saint-Venant equations and a simple flow calculation while demonstrating the LaTeX techniques I practiced.

1 Why HEC-RAS Interests Me

As a graduate in water resource engineering, I enjoy learning how water moves through rivers and drainage systems. **HEC-RAS** helps me explore how the geometry of the channel, flow resistance, and downstream water levels affect floods.

My undergrad thesis was based on *flood inundation modeling*. However, producing a flood map is only part of the job. Understanding the physics helps me interpret the results.

2 The Saint-Venant Equations

I tried to cover implementation of some basic agendas including equations that I learnt from the provided introductory tutorial.

2.1 Conservation of Mass

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0 \tag{1}$$

Equation 1 states that differences between inflow and outflow cause water storage to change.

2.2 Conservation of Momentum

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q^2}{A}\right) + gA\left(\frac{\partial z}{\partial x} + S_f\right) = 0 \tag{2}$$

Equation 2 describes changes in discharge, momentum transport, water-surface slope, and friction. Here, z represents water-surface elevation.

What I Practiced

I tried to apply some of the basic LaTeX features I learned from the provided introductory tutorial, including formatting equations..